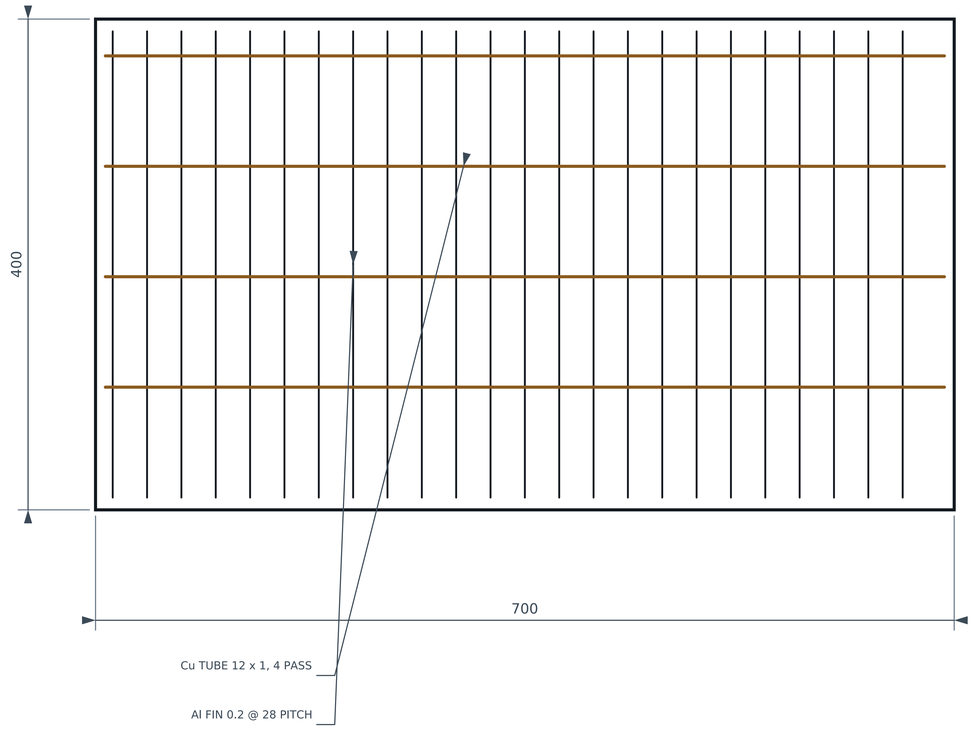
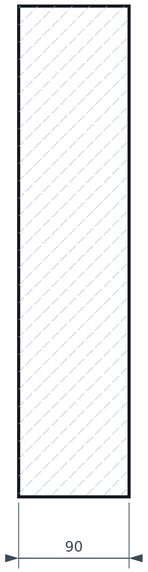
CONDENSER — FRONT

SCALE 1:5

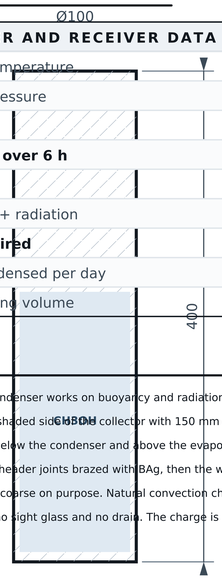


SIDE

1:5



RECEIVER SI-103



CONDENSER AND RECEIVER DATA	
condensing temperature	30 C
condensing pressure	21.9 kPa abs
duty per day	3.76 MJ
mean power over 6 h	**174 W**
air-side dT	15 K
h, convection + radiation	10 W/m2K
surface required	**1.16 m2**
methanol condensed per day	3.19 kg
receiver working volume	3.0 L

NOTES

1. No fan. The condenser works on buoyancy and radiation, so it must be shaded from the sun and free to draw air from below.
2. Mount on the shaded side of the collector with 150 mm clear beneath and above.
3. Receiver sits below the condenser and above the evaporator: the loop is gravity-driven, there is no pump.
4. Cu tube to SS header joints brazed with BAg, then the whole assembly TIG welded into the envelope.
5. Fin pitch 28 is coarse on purpose. Natural convection chokes below about 6 mm free spacing.
6. Receiver has no sight glass and no drain. The charge is not serviceable in the field.

WORKING LEVEL 3.2 kg

INSPECTION		
CHECK	**REQUIREMENT**	**METHOD**
Helium leak, whole assembly	< 1e-8 Pa.m3/s	mass spectrometer
Proof pressure	21 bar, 5 min	hydrostatic, then dry
Condenser orientation	tubes horizontal, drain to receiver	level check
Shading	no direct sun on the fins	site survey

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **CONDENSER AND RECEIVER**
AIR-COOLED, NATURAL CONVECTION / 174 W MEAN / 1.16 m2

SCALE: **1:5** — THIRD ANGLE PROJECTION
UNITS: **mm**

DRAWING No.: **SI-102 / SI-103**
REV: **A**
SHEET: **3 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15